Asia Pacific®
Institute of Management
Ahmedabad

A

Grand Project On

VEIL

Submitted to

Gujarat University

Faculty of Computer Applications

In partial fulfilment of the requirement of

the award for the degree of

Bachelor of Computer Application

GUJARAT UNIVERSITY

Gujarat University

Submitted by

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1. COLLEGE CERTIFICATE

2. ACKNOWLEDGEMENT

"The success of any person is not only his hard work, but it is also the support and co-operation one gets for success."

Apart from individual effort, the success of any project largely depends on the encouragement and guidance of many others. We take this opportunity to express our sincere gratitude to all those who have been instrumental in the successful completion of this project.

We express our deep gratitude to everyone who gave their valuable time by filling out our survey and helping us complete this project successfully. We would also like to thank the Principal I/C for providing a peaceful and supportive environment for our studies and for helping us in preparing this report.

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4. INTRODUCTION

4.1 Problem Definition

Phishing is a form of social engineering in which an attacker attempts to deceive a person into revealing sensitive information, opening unsafe content or visiting a fraudulent website. Such attempts may be delivered through website URLs, email messages and SMS messages.

VEIL studies whether machine-learning methods can classify inputs as legitimate or suspicious using information from these three channels. URL records contain structured website features, whereas email and SMS records are mainly text. Therefore, the project uses two data groups: a URL dataset and a combined email/SMS dataset, with separate preprocessing pipelines.

4.2 Objectives of the Project

- Study phishing-related signals in URLs, emails and SMS messages.
- Prepare and analyse URL data separately from combined email/SMS text data.
- Perform preprocessing, EDA and visualization using Python libraries.
- Train and compare three supervised classification algorithms.
- Evaluate models using suitable classification metrics and confusion matrices.
- Implement at least one advanced technique: K-Means, DBSCAN or Apriori.
- Interpret results and document limitations and future improvements.

4.3 Scope

VEIL is an academic prototype, not a production security service. The URL and email/SMS models are evaluated separately because their features and potentially their label definitions differ. SMS spam must not be described as confirmed phishing unless the dataset labels support that interpretation.

5. DATASET SELECTION

The course guideline requires real datasets from sources such as Kaggle or the UCI Machine Learning Repository. Record the exact source, licence, access date, number of samples, feature columns and target variable for every dataset actually used.

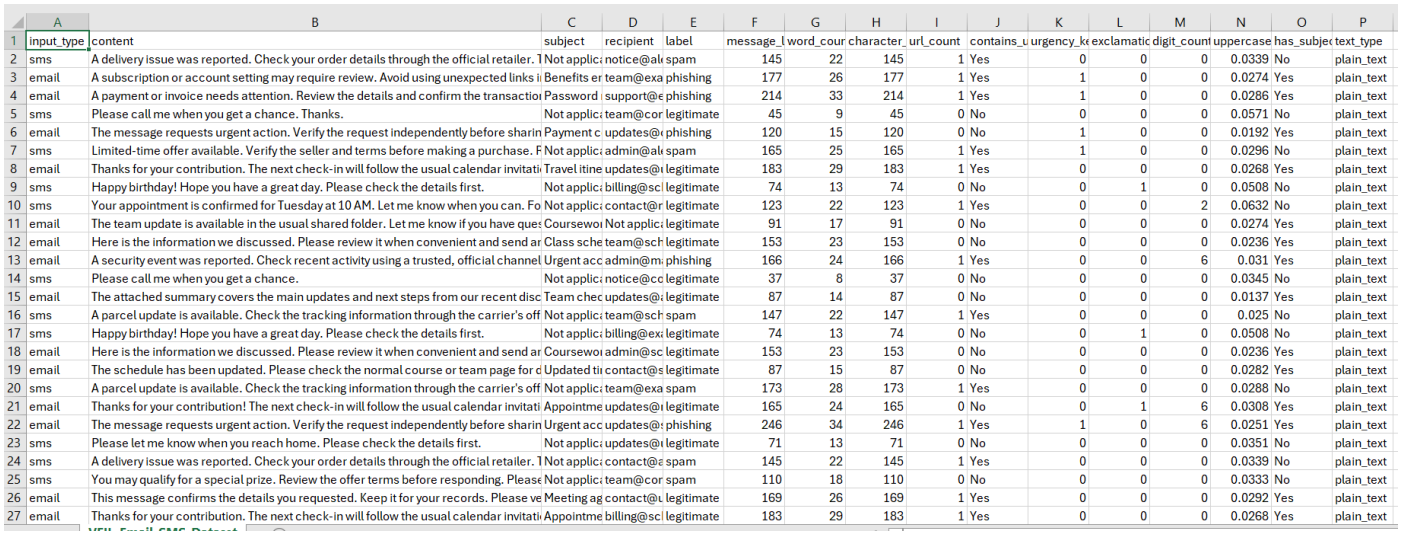
5.1 Dataset Group A — URL

The available URL workbook was supplied for this project. A prepared export named VEIL_URL_Dataset.xlsx has 11,590 rows, including 160 resampled duplicate observations; these are not 160 new independent records. The original cleaned workbook was previously observed with 11,430 rows. Verify the final file, unique count, provenance and target before submission.

	A	B	C	D	E	F	AQ	AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC
1	url	length_url	gth_hostna	ip	nb_dots	nb_hyphen	est_word_	est_words	est_word_	est_word_	g_words_r	g_word_h	g_word_p	phish_hint	main_in_br	l_in_subdc	and_in_pa	specious_	status
2	http://ww	37	19	0	3	0	3	11	11	6	5.75	7	4.5	0	0	0	0	0	legitimate
3	http://sha	77	23	1	1	0	2	32	19	32	15.75	19	14.66667	0	0	0	0	0	phishing
4	https://su	126	50	1	4	1	2	17	13	17	8.25	8.4	8.142857	0	0	0	0	0	phishing
5	http://rgip	18	11	0	2	0	0	5	5	0	5	5	0	0	0	0	0	0	legitimate
6	http://ww	55	15	0	2	2	4	11	7	11	6.333333	5	7	0	0	0	0	0	legitimate
7	http://app	32	24	0	3	1	0	7	7	0	4.5	4.5	0	0	0	0	0	0	phishing
8	http://ww	19	12	0	2	0	0	5	5	0	4	4	0	0	0	0	0	0	legitimate
9	http://ww	81	27	1	2	0	2	32	19	32	13.2	11	14.66667	0	0	0	0	0	phishing
10	http://van	42	34	0	2	0	0	21	21	0	14.5	14.5	0	0	1	0	0	0	legitimate
11	https://pa	104	10	0	1	10	2	10	6	10	5.571429	6	5.538462	1	0	0	0	0	legitimate
12	https://wv	56	22	0	3	0	3	15	15	7	6.666667	9	5.5	0	0	0	0	0	legitimate
13	https://wv	43	16	0	2	3	2	8	8	6	4.333333	5.5	3.75	0	0	0	0	0	legitimate
14	https://tec	83	14	0	1	9	3	10	10	10	5.454545	10	5	0	0	0	0	0	legitimate
15	http://htm	31	10	0	2	0	4	8	4	8	5.333333	4	6	0	0	0	0	0	phishing
16	https://wv	25	16	0	2	0	0	8	8	0	5.5	5.5	0	0	0	0	0	0	legitimate
17	http://wav	51	23	0	3	0	5	12	12	7	6.6	8	5.666667	0	0	0	0	0	phishing
18	https://wv	31	22	0	2	0	0	14	14	0	8.5	8.5	0	0	0	0	0	0	legitimate
19	http://beta	50	21	0	2	0	5	12	12	8	7	8	6.333333	0	0	0	0	0	phishing
20	http://ww	34	23	0	2	0	2	15	15	2	6.666667	9	2	0	0	0	0	0	phishing
21	http://ww	27	19	0	2	0	0	11	11	0	7	7	0	0	0	0	0	0	legitimate
22	http://ww	63	14	0	3	2	2	8	4	8	5.25	3.5	5.833333	0	0	0	0	0	legitimate
23	https://blo	66	16	0	2	4	4	9	7	9	5.875	5.5	6	0	0	0	0	0	legitimate
24	http://bat	40	11	0	2	0	1	7	7	7	5	7	4.5	0	0	0	0	0	phishing
25	http://sop	48	16	0	1	3	3	9	6	9	5.333333	5.5	5.25	0	0	0	0	0	legitimate
26	http://sup	126	50	1	4	1	2	17	13	17	7.769231	8.4	7.375	0	0	0	0	0	phishing
27	https://po	41	32	0	2	0	0	19	19	0	13.5	13.5	0	0	1	0	0	0	phishing

5.2 Dataset Group B — Combined Email and SMS

The current VEIL_Email_SMS_Mixed.xlsx file contains 23,180 AI-generated synthetic demonstration records (11,590 email and 11,590 SMS). These are not Kaggle/UCI records and do not meet the real-dataset requirement. The SMS label spam_unverified is not the same as confirmed phishing.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	input_type	content	subject	recipient	label	message_l	word_cour	character_	url_count	contains_u	urgency_k	exclamatic	digit_count	uppercase	has_subje	text_type
2	sms	A delivery issue was reported. Check your order details through the official retailer. T	Not applic	notice@al	spam	145	22	145	1	Yes	0	0	0	0.0339	No	plain_text
3	email	A subscription or account setting may require review. Avoid using unexpected links i	Benefits er	team@exa	phishing	177	26	177	1	Yes	1	0	0	0.0274	Yes	plain_text
4	email	A payment or invoice needs attention. Review the details and confirm the transactio	Password	support@e	phishing	214	33	214	1	Yes	1	0	0	0.0286	Yes	plain_text
5	sms	Please call me when you get a chance. Thanks.	Not applic	team@cor	legitimate	45	9	45	0	No	0	0	0	0.0571	No	plain_text
6	email	The message requests urgent action. Verify the request independently before sharin	Payment c	updates@	phishing	120	15	120	0	No	1	0	0	0.0192	Yes	plain_text
7	sms	Limited-time offer available. Verify the seller and terms before making a purchase. F	Not applic	admin@al	spam	165	25	165	1	Yes	1	0	0	0.0296	No	plain_text
8	email	Thanks for your contribution. The next check-in will follow the usual calendar invitati	Travel itine	updates@	legitimate	183	29	183	1	Yes	0	0	0	0.0268	Yes	plain_text
9	sms	Happy birthday! Hope you have a great day. Please check the details first.	Not applic	billing@sc	legitimate	74	13	74	0	No	0	1	0	0.0508	No	plain_text
10	sms	Your appointment is confirmed for Tuesday at 10 AM. Let me know when you can. Fo	Not applic	contact@r	legitimate	123	22	123	1	Yes	0	0	2	0.0632	No	plain_text
11	email	The team update is available in the usual shared folder. Let me know if you have ques	Coursewo	Not applic	legitimate	91	17	91	0	No	0	0	0	0.0274	Yes	plain_text
12	email	Here is the information we discussed. Please review it when convenient and send ar	Class sche	team@sch	legitimate	153	23	153	0	No	0	0	0	0.0236	Yes	plain_text
13	email	A security event was reported. Check recent activity using a trusted, official channel	Urgent acc	admin@m	phishing	166	24	166	1	Yes	0	0	6	0.031	Yes	plain_text
14	sms	Please call me when you get a chance.	Not applic	notice@cc	legitimate	37	8	37	0	No	0	0	0	0.0345	No	plain_text
15	email	The attached summary covers the main updates and next steps from our recent disc	Team chec	updates@	legitimate	87	14	87	0	No	0	0	0	0.0137	Yes	plain_text
16	sms	A parcel update is available. Check the tracking information through the carrier's off	Not applic	team@sch	spam	147	22	147	1	Yes	0	0	0	0.025	No	plain_text
17	sms	Happy birthday! Hope you have a great day. Please check the details first.	Not applic	billing@ex	legitimate	74	13	74	0	No	0	1	0	0.0508	No	plain_text
18	email	Here is the information we discussed. Please review it when convenient and send ar	Coursewo	admin@sc	legitimate	153	23	153	0	No	0	0	0	0.0236	Yes	plain_text
19	email	The schedule has been updated. Please check the normal course or team page for d	Updated ti	contact@s	legitimate	87	15	87	0	No	0	0	0	0.0282	Yes	plain_text
20	sms	A parcel update is available. Check the tracking information through the carrier's off	Not applic	team@exa	spam	173	28	173	1	Yes	0	0	0	0.0288	No	plain_text
21	email	Thanks for your contribution! The next check-in will follow the usual calendar invitati	Appointme	updates@	legitimate	165	24	165	0	No	0	1	6	0.0308	Yes	plain_text
22	email	The message requests urgent action. Verify the request independently before sharin	Urgent acc	updates@	phishing	246	34	246	1	Yes	1	0	6	0.0251	Yes	plain_text
23	sms	Please let me know when you reach home. Please check the details first.	Not applic	updates@	legitimate	71	13	71	0	No	0	0	0	0.0351	No	plain_text
24	sms	A delivery issue was reported. Check your order details through the official retailer. T	Not applic	contact@e	spam	145	22	145	1	Yes	0	0	0	0.0339	No	plain_text
25	sms	You may qualify for a special prize. Review the offer terms before responding. Please	Not applic	team@cor	spam	110	18	110	0	No	0	0	0	0.0333	No	plain_text
26	email	This message confirms the details you requested. Keep it for your records. Please ve	Meeting ag	contact@	legitimate	169	26	169	1	Yes	0	0	0	0.0292	Yes	plain_text
27	email	Thanks for your contribution. The next check-in will follow the usual calendar invitati	Appointme	billing@sc	legitimate	183	29	183	1	Yes	0	0	0	0.0268	Yes	plain_text

6. DATA PREPROCESSING

6.1 URL Data

- Load the verified final file; inspect dimensions, columns and data types.
- Check missing values, exact duplicates, duplicate URLs and constant columns.
- Verify feature definitions; remove identifiers and any target-leakage columns.
- Encode target labels consistently; split into training and testing sets.
- Scale numerical features for scale-sensitive models using a training-only pipeline.

6.2 Email/SMS Text Data

- Use verified public email and SMS data in final experiments.
- Keep input type to distinguish email from SMS; create a documented common text field.
- Handle structural blanks separately from true missing values.
- Normalize and tokenize text; transform it using a documented method such as TF-IDF.
- Preserve original labels; do not silently convert spam into phishing.

6.3 Split and Leakage Prevention

Split the data before fitting imputers, scalers, vectorizers or feature-selection steps. Fit transformations only on training data, preferably through Scikit-learn Pipelines. Where repeated templates or duplicate records exist, consider group-aware splitting to reduce leakage.

7. EXPLORATORY DATA ANALYSIS (EDA)

7.1 Univariate Analysis

Examine one variable at a time. For URL data, inspect target-class frequencies and distributions of numerical and categorical features. For email/SMS data, inspect input_type, labels, message length, word count and URL count.

7.2 Bivariate Analysis

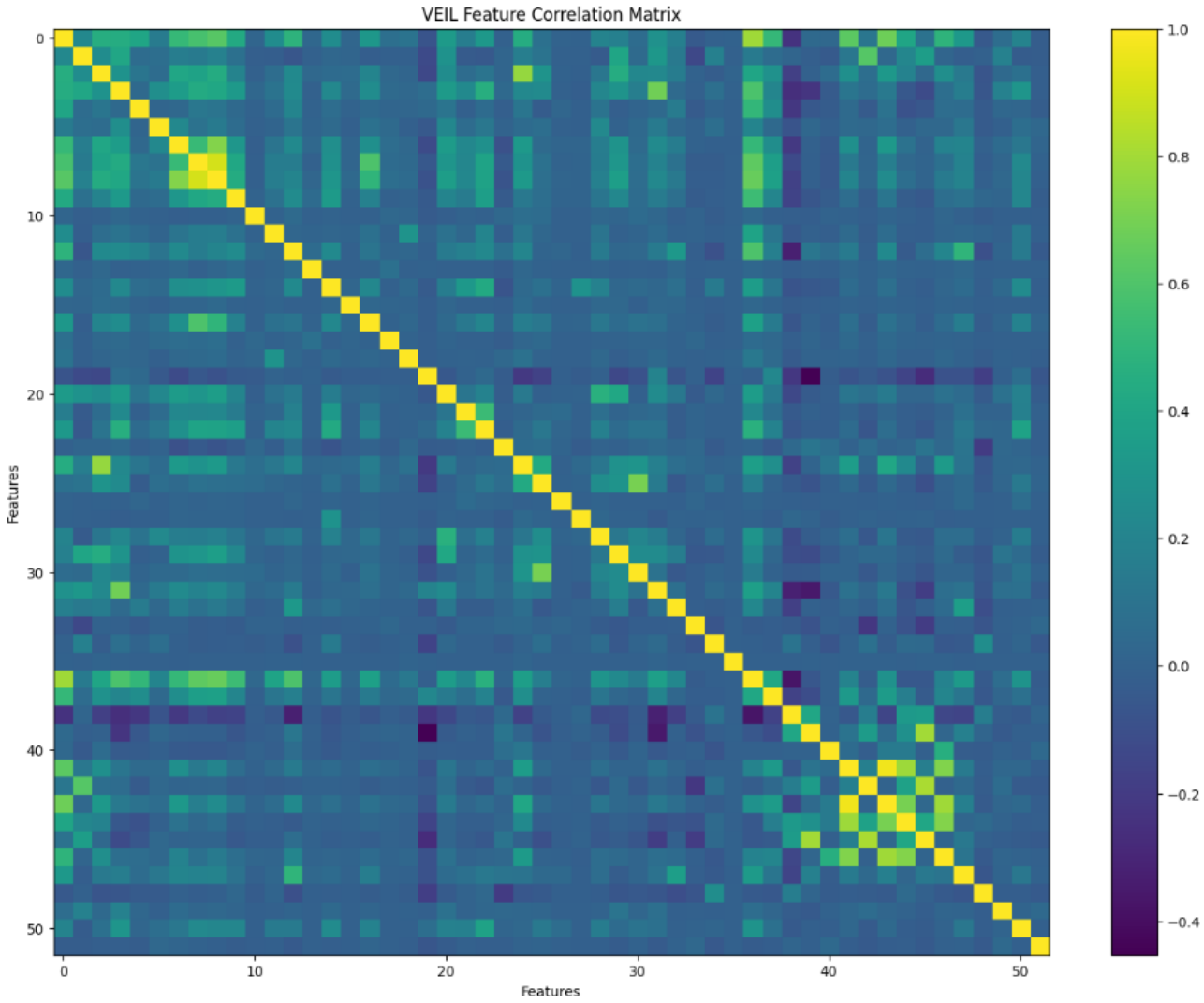
Examine relationships between pairs of variables, such as URL feature versus target, message length versus label, or input_type versus label.

7.3 Statistical Summary

Report count, mean, standard deviation, minimum, quartiles and maximum for suitable numerical features. Include class frequencies for categorical variables.

7.4 Correlation Analysis

Calculate correlations for suitable numerical features and explain the method used. Correlation does not prove causation. For high-dimensional text vectors, use a suitable feature-level analysis instead of an unreadable full matrix.


VEIL Feature Correlation Matrix
Features
Features
0
10
20
30
40
50
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

8. VISUALIZATION USING MATPLOTLIB AND SEABORN

Use Matplotlib and Seaborn to create readable figures. Each figure should include a figure number, descriptive title, axis labels, legend where needed and a short interpretation.

- Class distribution plot for the URL dataset.
- Histograms or boxplots for selected numerical URL features.
- Email/SMS input-type and label distribution plot.
- Message length or word-count comparison by class.
- Correlation heatmap for appropriate numerical features.
- Confusion-matrix heatmaps for the trained classifiers.
- Cluster or pattern visualization for the advanced technique.

MATPLOTLIB DIAGRAMS

1.Pie Chart – EMAIL-SMS

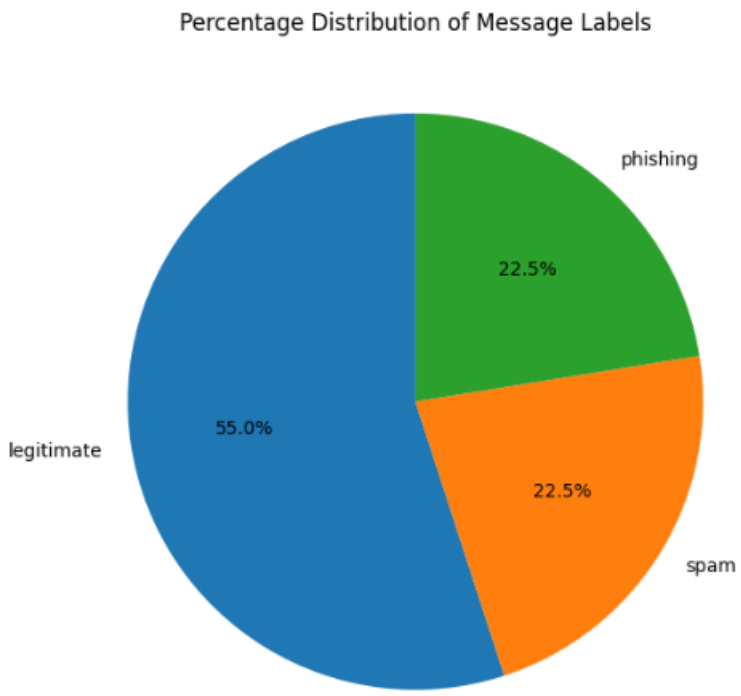
Message Category

```
[15]: label_percent = df['label'].value_counts(normalize=True) * 100

plt.figure(figsize=(7, 7))

plt.pie(
    label_percent,
    labels=label_percent.index,
    autopct='%1.1f%%',
    startangle=90
)

plt.title("Percentage Distribution of Message Labels")
plt.show()
```



SEABORN DIAGRAMS

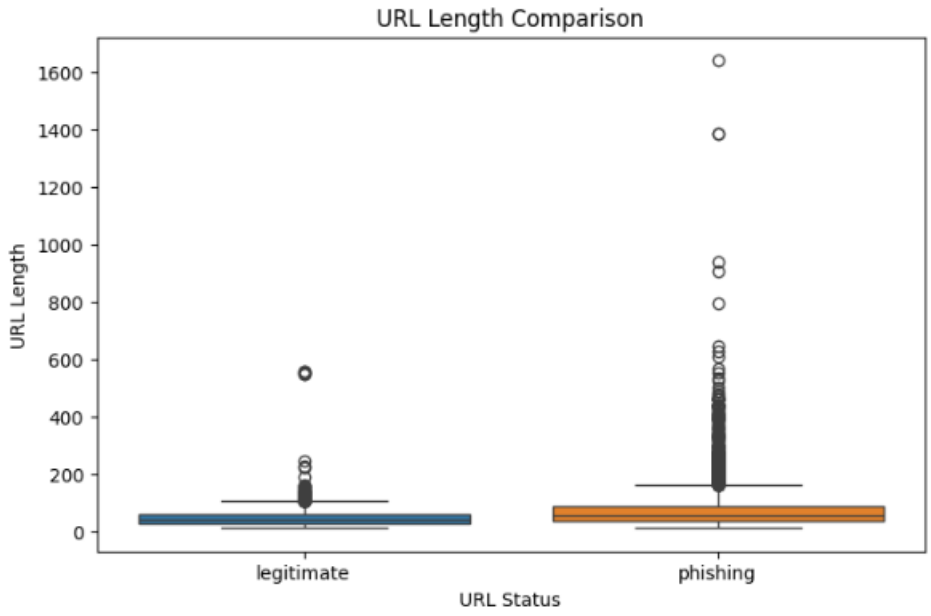
1. BOXPLOT- URL

```
[15]: plt.figure(figsize=(8, 5))

sns.boxplot(
    data=df,
    x="status",
    y="length_url",
    hue="status",
    legend=False
)

plt.title("URL Length Comparison")
plt.xlabel("URL Status")
plt.ylabel("URL Length")

plt.show()
```



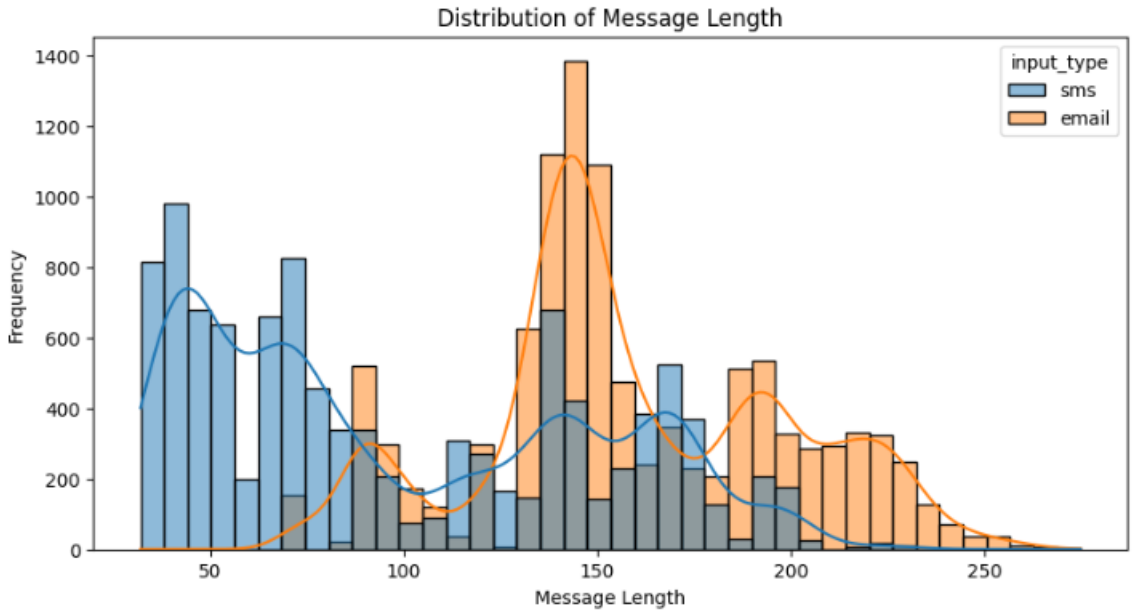
FREQUENCY DISTRIBUTION

1. EMAIL-SMS

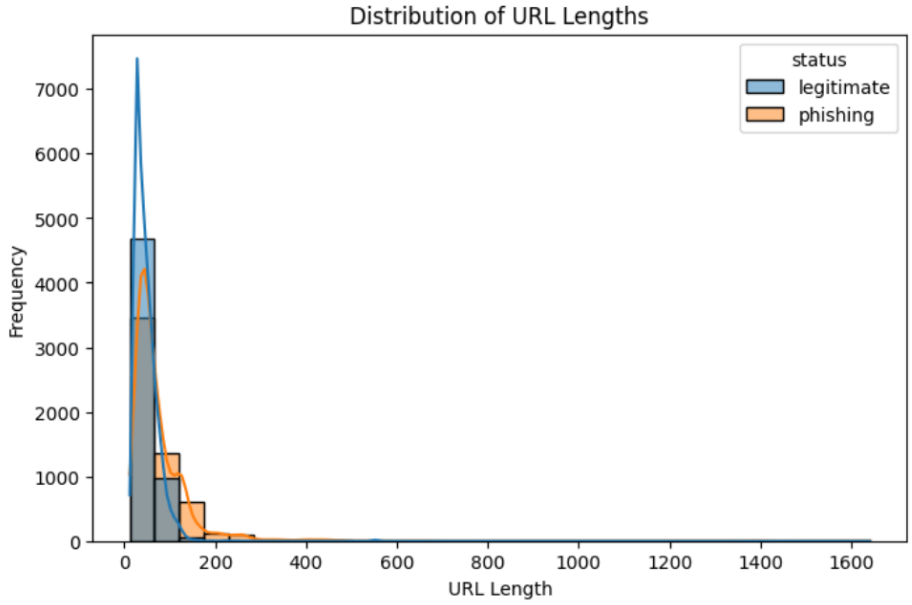
```
[18]: plt.figure(figsize=(10, 5))

sns.histplot(
    data=df,
    x='message_length',
    hue='input_type',
    bins=40,
    kde=True
)

plt.title("Distribution of Message Length")
plt.xlabel("Message Length")
plt.ylabel("Frequency")
plt.show()
```

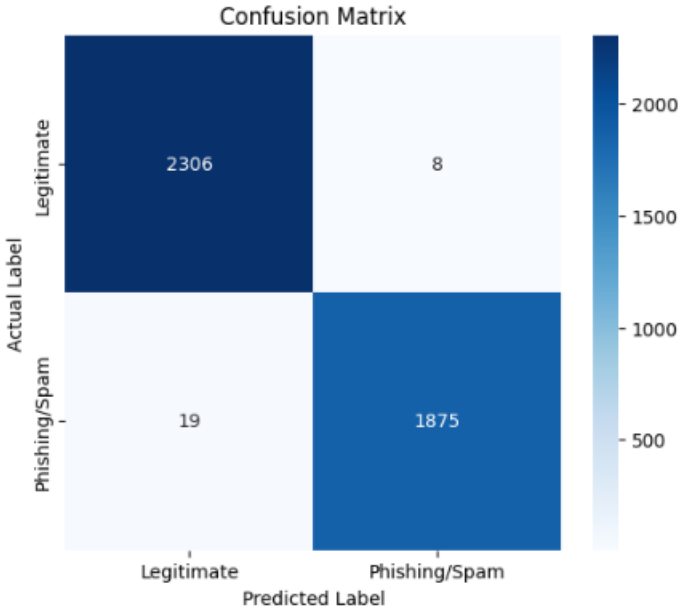


2. URL

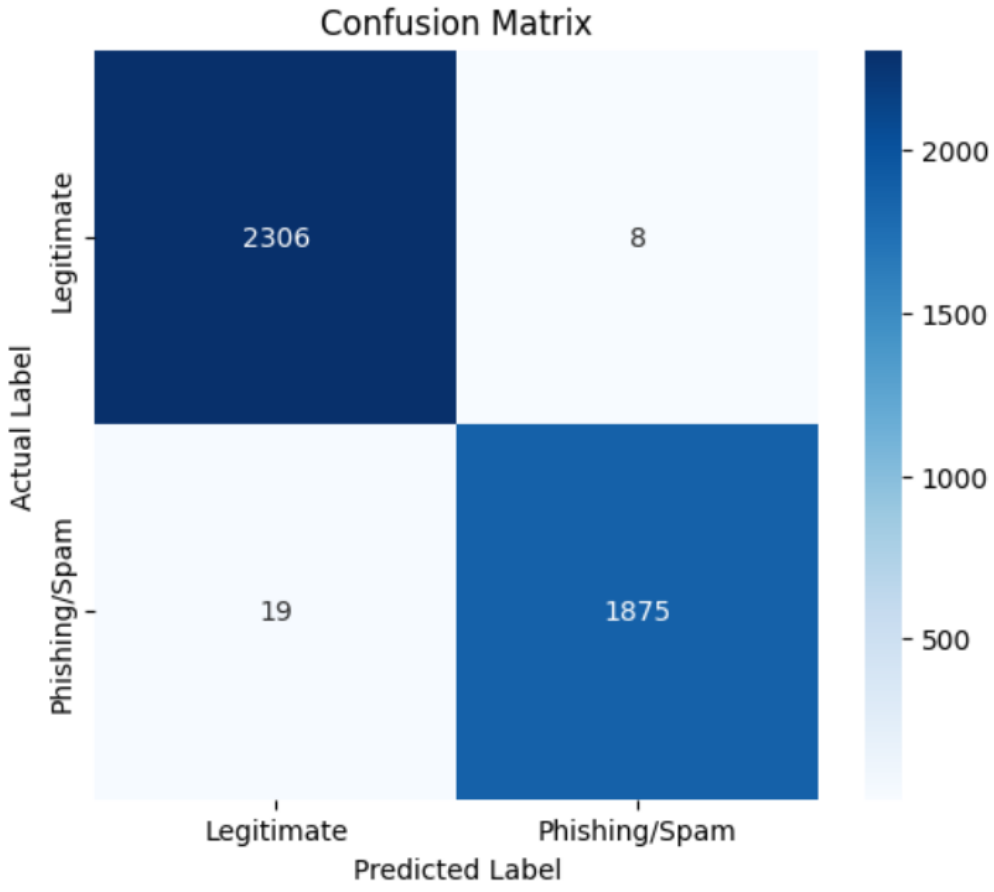


CONFUSION MATRIX

1.EMAIL-SMS



3. URL



9. MODEL EVALUATION AND PERFORMANCE METRICS

9.1 Problem Type

The intended task is supervised classification. State whether each dataset uses binary or multiclass labels after inspecting the verified target column. Do not treat SMS spam as confirmed phishing without supporting labels.

9.2 Metrics

Accuracy is the fraction of correct predictions. Precision measures how many predicted positives are truly positive. Recall measures how many actual positives are detected. F1-score is the harmonic mean of precision and recall. ROC-AUC measures ranking discrimination when suitable class probabilities or scores are available.

9.3 Train-Test Split and Confusion Matrix

Use a documented train/test split, preferably stratified when appropriate. Generate a labelled confusion matrix for each model and discuss true positives, true negatives, false positives and false negatives.

9.4 Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	99.3%	99.58%	99.00%	99.29%	N/A
Model 2					
Model 3					

10. MODEL BUILDING

Implement any three algorithms permitted by the guideline. A practical classification choice is K-Nearest Neighbours (KNN), Naive Bayes and Decision Tree. If SVM is chosen instead, document its settings and preprocessing. Linear Regression is not a standard classification algorithm.

10.1 K-Nearest Neighbours (KNN)

KNN predicts a class using nearby training observations. Scaling is generally important because distance calculations depend on feature magnitudes.

10.2 Naive Bayes

Naive Bayes is a probabilistic classifier based on Bayes' theorem. Select a variant suitable for the data representation; Multinomial Naive Bayes is commonly used with non-negative text features.

10.3 Decision Tree

A Decision Tree learns feature-based splits and can model non-linear relationships. Control tree complexity to reduce overfitting.

10.4 Training, Tuning and Comparison

Train and test all models using the same split protocol. Perform basic hyperparameter tuning using cross-validation on training data. Compare the held-out results and justify the best model using the project's priority, such as phishing-class recall while monitoring false positives.

11. ADVANCED TECHNIQUES AND PATTERN DISCOVERY

The guideline requires implementation of at least one of K-Means, DBSCAN or Apriori. The selected method must be executed on suitable data, visualized and interpreted. The section below is a plan until actual outputs are inserted.

11.1 Selected Technique — K-Means (Proposed)

K-Means groups numerical feature vectors into K clusters by assigning observations to the nearest centroid. Scale suitable features and justify K using evidence such as an elbow plot or silhouette score. If text features are used, document the vectorization and any dimensionality reduction.

11.2 Cluster / Pattern Visualization

11.3 Interpretation

12. CONCLUSION & FUTURE WORK

12.1 Conclusion

VEIL presents an academic approach to analysing suspicious content across URLs, emails and SMS. The use of a separate URL pipeline and a combined email/SMS text pipeline reflects the different structures of these inputs. The final conclusion must be based on verified datasets and actual preprocessing, EDA, model evaluation and advanced-technique results.

12.2 Limitations

- The currently available email/SMS examples are synthetic and must be replaced for the real-dataset requirement.
- The URL workbook's original provenance and exact unique sample count must be verified.
- SMS spam and phishing are not interchangeable labels.
- Class imbalance, duplicates, data leakage and dataset bias may affect performance.

12.3 Future Work

- Use verified, licensed public datasets and preserve source attribution.
- Evaluate on independent and more recent data.
- Improve explainability and error analysis.
- Build a user interface routing each input type to its appropriate model.

13. BIBLIOGRAPHY

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8. Gujarat University. *Bachelor of Computer Applications (BCA) Project Guidelines*.
9. Dataset sources: The original sources of the URL, email and SMS datasets used in the VEIL project.